

ANTOFAGASTA, Chile

WMO: 854420

Lat: 23.4503S

Lon: 70.4411W

Elev: 371

StdP: 14.50

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	50.0	51.7	42.7	40.9	54.1	44.3	43.6	55.1	21.6	58.3	19.4	58.6	5.5	70	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	75.6	66.1	74.9	65.5	73.5	64.6	68.4	73.7	66.9	72.5	65.6	71.5	12.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.9	72.4	64.5	92.1	70.4	63.0	87.2	69.1	32.9	73.8	31.7	72.7	30.7	71.8	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.1	17.8	16.4	DB	46.6	80.5	1.8	2.8	45.3	82.5	44.2	84.1	43.1	85.7	41.8	87.7	(4)
(5)			WB	42.9	68.8	2.7	1.7	41.0	70.0	39.4	71.0	37.9	71.9	36.0	73.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	68.7	68.6	66.7	63.1	60.4	58.2	57.1	57.9	58.9	60.7	63.2	65.9	(6)
(7)		DBStd	4.58	2.25	1.98	2.12	2.39	2.68	2.48	2.36	2.20	2.09	1.80	1.72	2.13	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	1279	0	0	6	67	146	205	244	221	184	134	59	13	(9)
(10)		CDD50	4528	579	521	517	392	322	245	221	245	267	332	395	492	(10)
(11)		CDD65	331	114	102	58	9	3	0	0	0	0	4	41		(11)
(12)		CDH74	202	97	66	23	2	3	0	0	1	0	0	0	10	(12)
(13)		CDH80	5	3	1	1	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.1	8.4	7.9	7.6	7.2	6.8	7.1	7.6	8.4	9.1	9.4	9.1	8.9	(14)
(15)	Precipitation	PrecAvg	0.9	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.4	0.2	0.0	0.0	0.0	(15)
(16)		PrecMax	5.6	0.2	0.0	0.2	0.0	0.0	0.5	1.0	3.6	2.0	0.6	0.5	0.2	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.3	0.0	0.0	0.0	0.0	0.0	0.2	0.3	0.8	0.5	0.1	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	78.0	76.5	73.1	71.6	68.2	66.5	67.7	67.5	68.8	71.1	75.4	(19)
MCWB			68.5	67.6	65.8	62.6	61.2	59.8	58.0	58.8	58.9	59.1	61.4	65.6	(20)	
2%		DB	76.6	75.5	73.8	70.5	68.1	65.5	64.0	64.2	64.7	66.4	69.4	73.2	(21)	
		MCWB	67.5	66.0	64.6	62.0	60.3	57.8	56.7	57.1	57.2	58.1	60.7	63.6	(22)	
5%		DB	74.9	74.8	73.0	69.4	66.2	63.4	62.3	62.3	63.0	64.8	68.1	71.7	(23)	
		MCWB	65.9	65.5	64.3	61.5	58.7	56.8	55.8	55.7	56.2	57.2	59.8	62.8	(24)	
10%		DB	73.4	73.5	71.6	67.9	64.6	62.4	60.8	60.9	62.3	64.2	66.8	70.2	(25)	
		MCWB	64.8	64.6	63.5	60.5	57.8	56.2	54.9	55.0	55.8	56.9	59.0	62.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.9	69.7	68.0	64.6	63.4	61.6	59.8	61.2	61.0	60.4	62.8	67.1	(27)
MCDWB			76.4	74.9	73.4	69.8	68.8	66.1	64.2	65.0	64.8	65.9	68.6	73.2	(28)	
2%		WB	69.2	68.0	66.6	63.3	61.2	59.4	57.7	58.2	58.8	59.2	61.5	65.0	(29)	
		MCDWB	74.6	73.3	71.7	68.3	66.1	63.7	62.3	62.8	62.8	64.5	67.2	71.1	(30)	
5%		WB	67.2	66.7	65.3	62.3	59.9	57.6	56.2	56.2	57.3	58.3	60.6	63.7	(31)	
		MCDWB	72.6	72.4	70.7	67.3	64.5	62.0	60.7	60.9	62.0	63.6	66.2	69.9	(32)	
10%		WB	65.6	65.6	64.3	61.4	58.9	56.6	55.3	55.2	56.3	57.5	59.8	62.8	(33)	
		MCDWB	71.6	71.5	70.0	66.3	63.4	61.1	59.7	59.8	61.0	62.8	65.4	69.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.3	10.5	9.8	9.0	8.3	7.8	7.3	7.2	7.4	8.1	9.2	(35)
MCDBR			10.6	10.8	11.1	11.0	10.1	9.6	9.3	8.6	7.9	8.2	8.9	9.9	(36)	
MCWBR			5.4	5.2	5.3	5.6	5.3	5.2	5.1	4.4	4.2	4.1	4.3	4.8	(37)	
5% WB		MCDBR	9.9	10.0	10.5	10.1	10.0	9.4	9.0	8.3	7.8	8.1	8.5	9.5	(38)	
		MCWBR	5.6	5.1	5.4	5.6	5.8	5.9	5.5	4.7	4.4	4.3	4.4	5.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.399	0.390	0.368	0.360	0.345	0.334	0.337	0.336	0.346	0.341	0.341	0.367	(40)	
taud		2.339	2.376	2.430	2.420	2.436	2.455	2.446	2.445	2.396	2.418	2.443	2.379	(41)		
Ebn at Noon		300	298	296	285	276	272	276	288	298	310	316	310	(42)		
Edh at Noon		43	41	37	34	31	29	30	33	37	39	39	42	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2708	2534	2210	1770	1345	1133	1202	1484	1921	2305	2614	2785	(44)	
(45)		RadStd	42	76	72	75	70	45	64	47	67	84	86	74	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page